Thermal Removal Method for Chang'e-5 Multispectral Data Using a Heat Conduction Model and Real Terrain

Jinning Li, Meizhu Wang, Bing Wu, Rui Xu, and Zhiping He

***Abstract*— The lunar mineral spectrometer (LMS) onboard the Chang'e-5—China's lunar exploration mission—collected in situ multispectral data up to 3 μm over a large area of the lunar surface for the first time. As Chang'e-5 operated near the Moon's noon local time, the spectral data are influenced by thermal radiation. In particular, the LMS multispectral data are in the form of characteristic spectral bands with discontinuities that make it difficult to directly apply the existing empirical models for thermal removal. Here, we present a thermal removal method based on the 1-D heat conduction model and real terrain data. A centimeter-scale temperature distribution map of the multispectral measurement region is calculated, which ranged from 330 to 362 K. We calculated the temperatures of four hyperspectral regions using an empirical model, compared them with those of the same region using the proposed method, and obtained a relative error of less than 1%. Afterward, we removed the effects of thermal radiation in the multispectral data using a thermal removal model with the temperature of the multispectral region. We compared multispectral data after thermal removal using our proposed method with hyperspectral data after thermal removal using an empirical model at similar locations. The results showed good consistency between the data corrected by both methods. This article proposed method does not depend on the setting of the spectral band, and it can provide a reference for thermal removal of Chang'e-5 multispectral data and that of subsequent in situ noncontinuous spectral detection data.**

***Index Terms*— Change'5, lunar mineral spectrometer (LMS), multispectral data, thermal removal.**

I. Introduction

CHINA'S lunar exploration mission Chang'e-5 successfully landed in the lunar northeast Oceanus Procellarum (43.05°N, 308.08°E) on 1 December 2020 and collected 1.731 kg of lunar soil samples [1], [2]. The lunar mineral spectrometer (LMS) was installed on the Chang'e-5 lander to acquire in situ spectra with the centimeter-scale spatial resolution of the lunar surface sampling area [3], [4], [5]. In situ spectral data up to 3 μm were acquired for the first time to support the study of $H_2O/-OH$ on the lunar surface. The LMS has two operating modes: hyperspectral and multispectral detection. In the hyperspectral mode, the LMS detects the region of interest in the sampling area of the lunar surface with a spectral range of 0.48–3.2 μm and a spectral interval of 5 nm. In the multispectral mode, the LMS adjusts the measurement position using a 2-D pointing mechanism, and a total of 180 positions in the sampling area of the lunar surface are spectrally measured, with 20 characteristic bands in the spectral range of 0.48–3.2 μm [6], [7]. The other main features of the LMS are shown in Table S1. The multispectral measurement mode can provide rapid spectral measurement over a larger area than the hyperspectral detection mode. This type of data provides a new perspective on mineral and $H_2O/-OH$ distribution and content of the study area, and the acquired data are of important research value.

However, because Chang'e-5 operated near the Moon's noon local time, the temperature of the lunar surface reached approximately 100 °C. Therefore, when the spectrum is greater than 2 μm, the impact of the measurement target's thermal radiation is not negligible [8], [9], [10], [11]. The detected spectrum not only comprises the spectrum of the target reflecting the Sun but also the target's thermal radiation spectrum. The reflection spectrum after 2 μm contains a wealth of information, particularly the $H_2O/-OH$ content, which will seriously affect the application of the spectral data if the thermal radiation component in the spectrum is not effectively removed.

Directly measuring the thermal radiation or temperature information of the detection area for thermal removal is an effective way. The Diviner onboard the lunar reconnaissance orbiter (LRO) has seven channels for measuring infrared radiation from 7.55 to 400 μm, which can be used to assess the temperature of the lunar surface [12], [13]. However, the difference in synchronization and spatial resolution between remote sensing and in situ instruments makes it difficult to use the temperature information measured by Diviner for the thermal removal of Chang'e-5 in situ data.

Indirect assessment of the lunar surface temperature is an effective way to obtain information about the target thermal radiation, and there are two main types of indirect assessment methods for the lunar surface temperature. One method is

Manuscript received 24 December 2022; revised 26 April 2023; accepted 22 May 2023. Date of publication 1 June 2023; date of current version 19 June 2023. This work was supported in part by the China Lunar Exploration Program (CLEP); in part by the China National Space Administration (CNSA); in part by the National Natural Science Foundation of China (NSFC) under Grant 21105109, Grant 61605231, Grant 62125505, and Grant 11941002; and in part by the Shanghai Outstanding Academic Leaders Plan. *(Corresponding authors: Rui Xu; Zhiping He.)*

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This article has supplementary downloadable material available at https://doi.org/10.1109/TGRS.2023.3281813, provided by the authors.

Digital Object Identifier 10.1109/TGRS.2023.3281813

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to assess the temperature of the measurement region using the empirical model. For example, to build an empirical model constrained by Diviner data, Li and Milliken [14] proposed an empirical thermal correction model for M^3 spectral correction using the observed reflectance correlations of 1.55 and 2.54 μm from the laboratory reflectance spectra of lunar soil samples collected back by Apollo and Luna missions. It can evaluate the temperature information of the measured target that is used to remove the influence of thermal radiation from the spectral data. This approach has now been successfully applied to the thermal removal of several lunar spectra [15], [16], [17], [18]. However, as empirical models are limited to specific spectral bands, it is usually necessary to measure continuous spectra and spectral bands that cover the spectrum unaffected by thermal radiation up to 2 μm. When the spectral measurements do not involve a specific spectral band, empirical models are difficult to apply.

Another method is based on the heat conduction model to estimate the temperature of the lunar surface. This has been widely used in the assessment of temperature properties of extraterrestrial bodies without atmosphere (Moon and asteroids) [19], [20], [21], [22]. Hayne et al. [23] used the heat conduction model, combined with a lunar surface roughness model, to estimate the temperature distribution of the lunar surface. Liu and Jin [24] constructed a real-time lunar surface temperature simulation model based on the heat conduction model to calculate the temperature variation in the permanently shadowed region of the Moon's South Pole. Bandfield et al. [25] estimated the temperature of the lunar surface based on a heat transfer model by constructing a model that simulates the weathering layer roughness of the lunar surface, thus eliminating the thermal contribution from the M^3 spectral data.

The multispectral mode of LMS acquires spectral data in noncontinuous spectral bands, rendering empirical models difficult to apply. Therefore, temperature assessment based on the heat conduction model is a feasible approach. However, the accuracy of this method for temperature assessment relies on obtaining relatively reliable lunar surface roughness [26], [27], [28].

Here, we propose an in situ multispectral data thermal removal method based on a heat conduction model combined with real terrain. This method does not depend on the setting of the spectral band, and only the precise topography of the measurement area and the moment of measurement are required to obtain the surface temperature of the Moon and achieve the thermal removal of the spectrum. The method is not only applicable to the thermal removal of Chang'e 5 multispectral data but can also be applied to the thermal removal of other in situ noncontinuous spectral data.

II. Microscale Temperature Distribution in the Lunar Surface Sampling Area

A. Temperature Assessment Method

The stereo surveillance camera (SSC) onboard Chang'e-5 has the same multispectral measurement area as the lunar mineralogical spectrometer [see Fig. 1(a)]. Based on the terrain

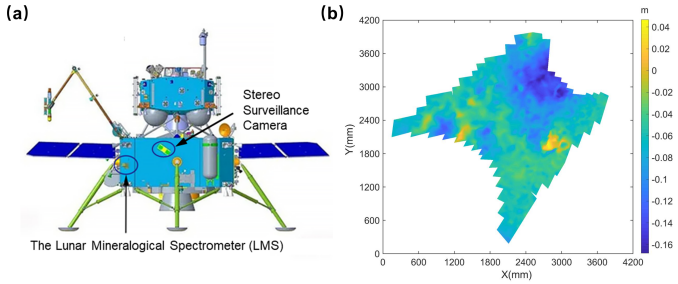


Fig. 1. Schematic of the lunar surface operating area and its DEM data for LMS onboard Chang'e-5. (a) Illustration of the installation position of LMS and SSC on Chang'e-5 lander. (b) DEM terrain distribution map of the multispectral detection area.

image data of the sampling area acquired by the SSC, a DEM image of the sampling area can be synthesized. Fig. 1(b) presents the DEM data of the multispectral measurement area with a data resolution of 3 mm/pixel.

Chang'e-5 operated on the lunar surface at approximately 10:00 local time. During the measurement of multispectral data, the solar incidence angle at the lunar surface did not change significantly (see Table S2); hence, the heat conduction model can be used to calculate the temperature at the lunar surface. The heat conduction model simulates the process of heat change in the lunar soil caused by solar radiation. The Sun's radiation heats the lunar soil, and part of the heat is transferred to the deep lunar soil through the contact between the lunar soil particles, whereas some of the energy is radiated to space through the surface. Considering that there is no visible deep crater in the Chang'e-5 spectral measurement area, the radiated energy from the secondary reflection from the lunar soil crater wall can be ignored. Thus, the temperature of the LMS lunar surface can be evaluated using the 1-D heat conduction model, and the 1-D heat conduction relation of the temperature T of the lunar soil with local time t and depth z can be expressed as

$$\rho c_p \frac{\partial T}{\partial t} = \frac{\partial}{\partial z}\left(K \frac{\partial T}{\partial z}\right) \tag{1}$$

where ρ is the density of the lunar soil layer in kg/m^3, c_p is the heat capacity of the lunar soil in Jkg^{-1}K^{-1}, and K is the thermal conductivity of the lunar soil in Wm^{-1}K^{-1}.

The specific numerical relationships of the above parameters can be found in the Text S1. The boundary conditions on the surface layer of the heat conduction equation can be expressed as

$$\begin{aligned} K(z,T)\frac{\partial T(z,t)}{\partial z}\bigg|_{z=0} &= \varepsilon\sigma T_0^4 - (1-A)F \\ A(\theta) &= 0.12 + 0.06\left(\theta/45^\circ\right)^3 + 0.25\left(\theta/90^\circ\right)^8 \\ F(\theta) &= \frac{S}{d^2} * (\cos\theta + |\cos\theta|) * 0.5 \end{aligned} \tag{2}$$

where σ is Stefan–Boltzmann's constant equal to 5.67×10^{-8}Wm^{-2}K^{-4}, T_0 is the temperature of the surface as a function of local time t, ε is the average infrared irradiance, A is the average albedo of the lunar surface as a function of the solar incidence angle, F is the heat flow incident to the lunar

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surface, S is the total solar radiation intensity at 1 AU, which is equal to 1367 W/m^2, and d is the distance between the Sun and the Moon [23]. Because the surface of the measurement area is mostly small-scale slopes, ε was set to 0.85 [29].

According to (2), only the solar incidence angle θ is an unknown variable. Once the solar incidence angle in the region is obtained, the temperature of the surface can be solved. The solar incidence angle θ is the local incidence angle that depends not only on the latitude and longitude of the subsolar point on the lunar surface (φ, γ) and the latitude and longitude of the probe region (φ_0, γ_0) but also on the normal vector of the lunar surface slope in the probe region on the lunar surface. The incidence angle can be expressed as

$$\theta = \arccos(\hat{\alpha} \cdot \hat{\beta}) \tag{3}$$

where $\hat{\alpha}$ is the unit vector of the lunar center pointing to the subsolar point that is related to the latitude and longitude of the subsolar point (the specific relationship can be found in Text S2); $\hat{\beta}$ is the normal vector of the slope of the lunar surface in the measurement area and can be calculated from the DEM data.

The infrared field-of-view detection area of Chang'e-5 LMS is of centimeter scale, and the resolution of DEM data is of millimeter scale. To match the resolution of the two and reduce the volume of calculated data, we downsampled the DEM data by a factor of 10 to obtain the DEM data with a resolution of 3 cm. We divided the DME data into small right triangle-shaped slopes, with side lengths of 3 × 3 cm, and calculated the unit normal vector $\hat{\beta}$ for each small slope (see Text S2). The latitude and longitude of the measurement area are known, and the latitude and longitude of the lunar subsolar point at the time of measurement can be calculated according to VSOP82 planetary theory and Chapront ELP-2000/82 lunar theory, combined with LMS data timestamp [30].

The above parameters are incorporated into the 1-D heat conduction model, and the surface temperature of each small slope in the measurement area was solved to obtain the temperature distribution of the entire area.

B. Temperature Distribution of the Chang'e-5 Multispectral Measurement Region

The temperature distribution map of Chang'e-5 LMS at the centimeter scale over the entire observed lunar soil area surface was calculated using a 1-D heat conduction model, combined with real terrain DEM data of the measurement area [see Fig. 2(a)]. The regions in and around the rocks of the lunar surface have been removed from the figure because the temperature distribution calculations using 1-D heat conduction models with physical parameters of the lunar soil are not applicable to the areas. The temperature distribution in the measurement area was not uniform owing to the microscale terrain undulations of the lunar surface. The overall temperature ranged between 330 and 362 K, with an average of 350 K and a variation of approximately 30 K on the sunny and shady sides.

To verify the accuracy of the temperature range calculated by our proposed method, we compared the temperature

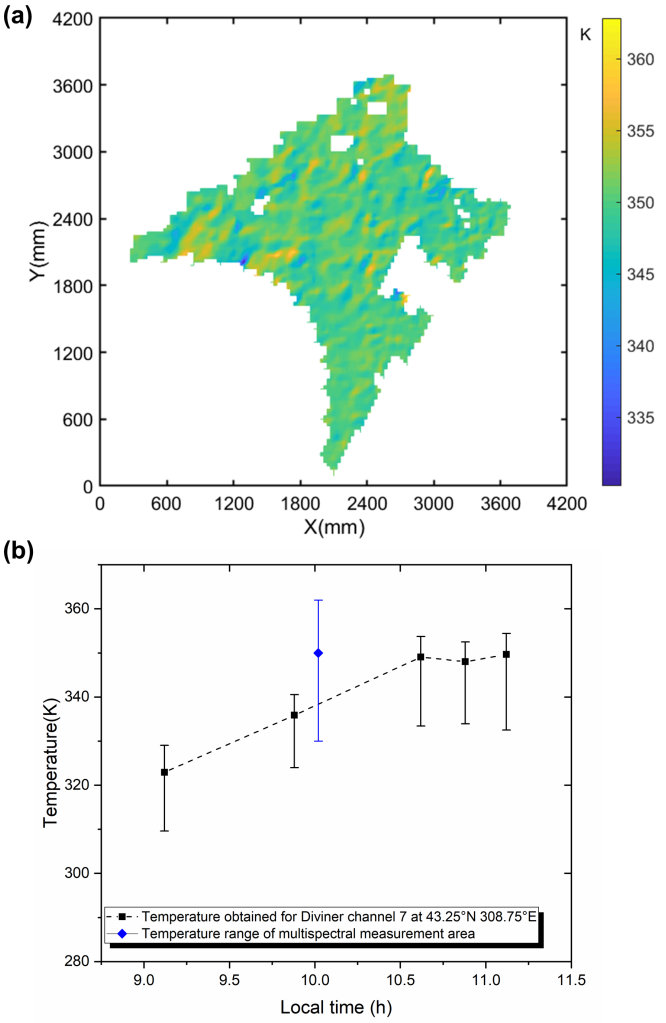


Fig. 2. Temperature map of the multispectral measurement area and temperature comparison. (a) Map of the temperature distribution in the multispectral measurement area with a spatial resolution of 3 cm/pixel. The blank area in the middle denotes the non-Moon soil part. (b) Comparison of the temperature variations obtained by Diviner channel 7 in the region at local time, and the results of the assessment of our proposed method.

measurements by the Diviner channel 7 at local time in a similar region (43.25°N, 308.75°E), as shown in Fig. 2(b) [31]. The temperature range distribution of Chang'e-5 multispectral area is similar to that of Diviner channel 7.

To further demonstrate the accuracy of the temperatures calculated by our proposed method, we calculated the temperatures of four hyperspectral detection regions measured by the LMS, all of which were measured on the lunar soil surface numbered D004, D009, D010, and D012, using the empirical model proposed by Li and Milliken [14]. The specific positions of the four areas are shown in Fig. S2. The average temperatures of the same areas were also calculated using our proposed method. The temperatures calculated using the two methods are in relatively better agreement, with relative errors below 1% (see Table I). The relatively large deviation in temperature obtained using the two methods at positions D009 and D012 may be related to the presence of some noise in the hyperspectral data used in the empirical model.

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TABLE I
COMPARISON OF THE TEMPERATURE OBTAINED BY THE TWO METHODS

Number	Empirical models (T_E)	This paper (T_P)	Relative error (T_E- T_P) / T_E*100%
D004	359.3K	359.4K	0.03
D009	349.6K	346.9K	0.77
D010	339.7K	339.8K	0.03
D012	343.5K	342.2K	0.38

III. THERMAL REMOVAL OF CHANG'E-5 MULTISPECTRAL DATA

A. Thermal Removal Method

The energy received by the LMS mainly consists of solar energy reflected by the target and energy emitted by the target as shown in the following:

$$I_{\text{meas}} = \frac{R_c F_{\text{sun}}}{d^2 \pi} + \varepsilon_{\text{IR}} I_e \tag{4}$$

where I_{meas} is the radiance obtained after LMS calibration, R_c is the actual reflectance of the target, F_{sun} is the irradiance of 1 AU of the Sun, ε_{IR} is the radiance of the infrared band with respect to the band, and I_e is the thermal radiation entering the instrument in the measurement field of view. Here, we assume that Kirchhoff's law holds, i.e., $\varepsilon_{\text{IR}} = (1 - R_c)$. Therefore, R_c can be expressed as

$$R_c = \frac{I_{\text{meas}} - I_e}{\frac{F_{\text{sun}}}{d^2 \pi} - I_e}. \tag{5}$$

There are several small slopes in the multispectral detection field of view, and each small slope has a different sloping angle. Thus, the average intensity of thermal radiation entering the field of view needs to be calculated as follows:

$$I_e = \frac{\sum_{i=1}^{N} Ir_i \cdot S_i \cdot \cos(\theta_{ei})}{\sum_{i=1}^{N} S_i \cdot \cos(\theta_{ei})}$$
$$Ir_i = \frac{c_1}{\lambda^5} \frac{1}{e^{c_2/\lambda T_i} - 1} \tag{6}$$

where θ_{ei} is the radiation angle of the ith slope; S_i is the area of the ith slope; I_{ri} is the thermal radiation intensity of the slope, related to λ being the spectral band and the temperature T_i of the slope; c_1 and c_2 are the first and second radiation constants, equal to $3.7415 \times 10^4 \text{Wcm}^{-2}\mu\text{m}^4$ and $1.43879 \times 10^4 \mu\text{mK}$, respectively.

The temperature T_i of the small slope can be calculated from I_{meas} obtained by LMS using the 1-D heat conduction model to obtain the reflectivity R_c, which is the reflectivity after the removal of thermal radiation.

B. Thermal Removal Result

At 10:03 local time, LMS carried out a multispectral survey of the sampling area on the lunar surface. The solar altitude angle and incidence angle at the time of the survey are listed in Table S2. In total, 180 areas were surveyed. Because the operation of Chang'e-5 sampling robotic arm during multispectral measurement of the detection area has a certain degree of obscuration, the data in some areas are affected to an extent.

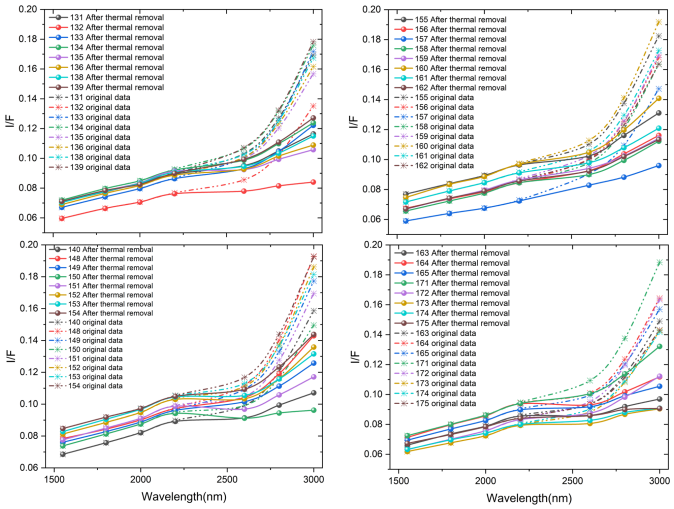


Fig. 3. Multispectral raw I/F emissivity data from 1550 to 3000 nm and I/F reflectance data after thermal removal.

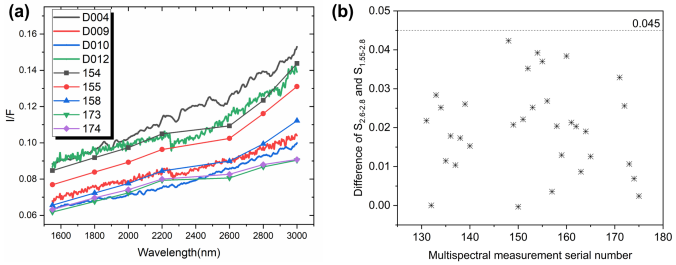


Fig. 4. Verification of thermal removal effect. (a) 154th, 155th, 158th, 173rd, and 174th multispectral data thermally corrected using our proposed method were compared with the D004, D009, D010, and D010 hyperspectral data corrected using the thermal removal method of Li and Milliken [14]. (b) Distribution of the difference between the slopes of 2.8–2.6 and 2.8–1.55 μm of the spectral data after thermal correction.

Moreover, the part close to the Chang'e-5 rover was affected by the engine plume and shading due to the sampling mechanical arm (see Fig. S1). Hence, when using it, we removed the data from the affected part, and the final multispectral mode still had 32 areas of detection data (see Fig. S2). Table S3 lists the 32 sets of data serial numbers. Based on the estimated lunar surface temperature distribution, the reflectance data of Chang'e-5 after hyperspectral and multispectral thermal removal were calculated using the thermal removal method described in Section II-A. Fig. 3 shows the multispectral raw I/F emissivity data for all 32 regions from 1550 to 3000 nm and the I/F reflectance data after thermal removal.

We used the method of Li and Milliken [14] to calculate the thermal removal of the D004, D009, D010, and D012 hyperspectral data and compared them with the five sets of thermally removed multispectral data [see Fig. 4(a)]. The multispectral acquisition positions numbered 154 and 155 were similar to hyperspectral position D004 and D012, while multispectral data numbered 158 were similar to hyperspectral position D009. Multispectral data with serial numbers 173 and 174 were also similar to hyperspectral position D010. The hyperspectral data for thermal removal using the empirical model were in general better consistency with the data for thermal removal obtained using our proposed method.

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To further determine the effect of spectral thermal removal, we used a thermal removal check method similar to that of Li and Milliken [14]. First, we determined whether the thermal removal was excessive by evaluating whether the spectrum showed a downward trend before 2.65 μm. Fig. 3 clearly shows that wavelengths less than 2.65 μm showed no downward trend after thermal correction, confirming that the thermal removal was not excessive. The difference between the slopes of 2.9–2.7 and 2.7–1.5 μm must be less than 0.2 to determine whether the heat removal is complete. Because the multispectral data for 2.9, 2.7, and 1.5 μm are not available, we determined whether the thermal removal was complete by calculating the difference between the slope of 2.8 and 2.6 μm and that of 2.8 and 1.55 μm [see Fig. 4(b)]. The differences in the slopes are all less than 0.045, indicating that the thermal effects in the multispectral data are effectively removed.

IV. Conclusion

A thermal removal method combining thermal conduction model and real terrain was adopted to realize the thermal removal of Chang'e-5 multispectral data. We confirmed the accuracy of the temperature assessment results by comparison with those detected by Diviner Channel 7 at the same local time in a similar area and the temperature results of the hyperspectral detection area calculated using an empirical model. Moreover, thermal removal of the LMS in situ multispectral data was performed based on the calculated detection area temperature. A comparison with the empirical model-corrected hyperspectral data and an examination of the thermal correction effect showed that the proposed method can effectively remove the thermal influence from the multispectral data.

The multispectral data after thermal removal using our method can provide data support for the research of water and hydroxyl and mineral composition content and distribution in the Chang'e-5 exploration area, and the obtained ground surface temperature distribution of the lunar surface provides a reference for the study of the microscale temperature distribution of the lunar surface and thermophysical properties of lunar soil. In the future, lunar exploration will become more refined, and more applications will be developed to detect the composition and content of certain specific substances using narrow or characteristic spectral bands. Our proposed method is independent on the setting of spectral bands and therefore represents a novel approach for the subsequent targeted thermal removal of spectral measurement data.

Acknowledgment

The authors would like to thank the National Astronomical Observatories, Chinese Academy of Sciences, Beijing, China, for the digital elevation model. They also thank Paul O. Hayne's team for open-sourcing the 1-D heat transfer modeling program.

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